

Sunny Zhang

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EDUCATION

University of Toronto, St. George

2025 – 2029

Honours BSc, Double Major in Computer Science and Mathematics

GPA: 3.7/4.0

PROJECTS

Feedback or Annealing? | *Python, NumPy, Numba, Matplotlib*

- Derived a geometric envelope for residual-driven aggregation from Bellman span contraction, creating an open-loop control that isolates feedback from the annealing schedule it induces.
- Evaluated matched residual, geometric, and fixed-width methods on a 9,261-state MDP across 3 update mixes, 20 paired seeds, and 5 compute budgets; feedback improved error by at most 0.3% and worsened it by up to 14.5%.
- Showed that annealing changed error by up to 23.5% relative to fixed resolution while all final feedback comparisons remained within a preregistered ± 0.02 equivalence region.

Parallel State Aggregation | *Python, NumPy, Numba, Matplotlib*

- Built a multithreaded benchmark of value iteration and adaptive state aggregation across 40,000–1,000,000-state MDPs, 1–10 threads, and paired seeds.
- Found value iteration reached the fixed ℓ_∞ error target 1.3–2.2 $\times$ faster across all 5 configurations; traced scaling limits to memory bandwidth and serial rebinning and exposed 99.67% uncounted work.

LaunchPilot | *Next.js, TypeScript, FastAPI, Postgres*

- Built an AI launch copilot in 72 hours with a 2-person team at Hack Canada 2026; won the sole Backboard.io Best Use prize (1 of 59 teams) and placed top 3 of 94 in the SPUR startup track.
- Designed a supervised multi-agent workflow converting project briefs or GitHub repositories into market research, ICPs, positioning, messaging, KPIs, execution plans, and outreach drafts.
- Orchestrated three 5-agent stages with persistent threads and memory snapshots, gated by GitHub context, Auth0-scoped permissions, and human approval before execution.

EXPERIENCE

LLM/ML Research Intern

Jun 2026 – Present

Algoverse AI Research Program

- Investigating whether hierarchical rank pressure causes pretrained LLM trading agents to misreport executions or withhold profit-improving information, using simulator-derived ground truth rather than LLM judges.
- Designing an ablation study across situational features, observable logs, private trader chat, and private reasoning; encoding each source as a fixed feature block and training regularized logistic-regression detectors to isolate the predictive value of private reasoning.
- Evaluating next-round misconduct prediction using AUPRC, AUROC, calibration, and recall under a fixed inspection budget, with run-level train/test splits and held-out pressure settings to measure out-of-distribution generalization.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL

Machine Learning and Data: PyTorch, Gymnasium, CleanRL, Pandas, NumPy, Numba

Frameworks: FastAPI, Next.js, Flask, D3

Tools and Platforms: Git, Docker, Supabase, Auth0, Hugging Face, pytest, Vercel